

ZULHELMI SYAHTIAR

Fresh Graduate D3 Industrial Electronics – IoT & Embedded Systems

Cipayung, Jakarta Timur | 0838-9446-9333 | helmi2591@gmail.com | [linkedin.com/in/zulhelmisyahtiar/](https://www.linkedin.com/in/zulhelmisyahtiar/) | [zulhelmi-portfolio](#)

SUMMARY

Fresh graduate in Electrical Engineering, D3 Industrial Electronics program. Strong interest in the Internet of Things (IoT), microprocessors, programming, and control systems. Experienced in microcontroller-based projects and the development of simple automation systems. Eager to keep learning, innovating, and applying the latest technology in the industrial world.

EXPERIENCE

PT PLN (Persero) ULTG Petukangan – Tangerang Selatan

Jul 2025 – Nov 2025

Protection Maintenance Intern (HAR PRO)

- Assisted with periodic maintenance and monitoring of substation protection systems.
- Performed testing of protection relays (OCR, GFR, Differential, Distance Relay, OVR/UVR, Buchholz, Thermal Overload).
- Involved in system testing during DSS migration at a 150kV substation.
- Carried out Battery Capacity Tests (BCT) to maintain the reliability of substation backup power.
- Assisted with non-cascade testing on transformer incoming panels.

Independent Clients – Jabodetabek

Mar 2023 – Present

Freelance Graphic Designer & Photographer

- Provided graduation and post-thesis-defense photography services, including style direction and photo editing.
- Created banner designs for graduation celebration events.
- Created social media graphic designs for organizational Instagram content.

EDUCATION

Politeknik Negeri Jakarta – Depok, Jawa Barat

2023 – 2026

D3 Industrial Electronics – GPA 3.55/4.00

- Final Project: "Dual DC Motor Control Trainer Kit – PID – Fuzzy Control Based on ESP32 with 3D Digital Twin Visualization".
- Teaching Assistant for a Community Service project installing a solar-powered Hydroponic Green House.
- Board Member of KSM PSYCHOROBOTIC, Communication & Information Department (2025).
- Committee member, Publication, Documentation & Creative Media Division at E-TIME 2025, Olahraga Elektro 2024, FRACTION 2024 (Electrical Engineering Department Graduation), and the Industrial Electronics Study Excursion.
- Committee member, Media Division at the EC-IKI PNJ 2024 Public Lecture.

SMAN 64 Jakarta – Jakarta Timur

2019 – 2022

High School – Science Major (MIPA)

CERTIFICATIONS

- BNSP Competency Certification – Instrumentation Technician (2026) – Competent. Competency test: zero-span configuration and scaling of a level transmitter (4–20 mA) on a PLC, including linearity and measurement-deviation analysis.
- TOEIC Prediction Test, Politeknik Negeri Jakarta (2026) – Score 620/990 (Listening 260, Reading 360).

FEATURED PROJECTS

- [DC Motor Control Trainer Kit](#) (2025–2026): Web monitoring dashboard (JavaScript ES6, HTML5, Tailwind CSS v4) and an interactive 3D Digital Twin (Three.js r160) synchronized in real time with ESP32 hardware via Firebase Realtime Database.
- IoT Web Monitoring Dashboard (2025): Monitoring of temperature and soil & air humidity based on a MySQL database.
- Plant Monitoring Robot (2025): ESP32-C3 with DHT11 and soil moisture sensors, IoT-based.
- Logic Gate Training Module (2024): Practice kit for AND, OR, NOT, NAND, NOR, XOR, Full Adder, Multiplexer, Demultiplexer, Flip-flop, and BCD-to-7-Segment Decoder.
- 9V Power Supply based on IC 7809 (2024): Design, assembly, and testing of a stable power supply for small-scale electronic devices.
- [Numa Space](#) (2026) A virtual study room and real-time study-buddy matchmaking platform. Enables P2P video/audio interaction via WebRTC, Socket.IO chat communication, a private room access code system, and Pomodoro Timer integration for productivity. Live Web

SKILLS

- IoT & Microcontrollers: ESP32, Arduino, Raspberry Pi, Firebase (Realtime Database, Firestore & Hosting).
- Control & Automation Systems: PID, Fuzzy Logic Controller, PLC programming, SCADA (Schneider Electric, Vijeo Citect).

- Protection & Testing Systems: Protection Relays (OCR, Differential, REF MiCOM), OMICRON TestUniverse, GOOSE Configuration.
- Programming: Python, C/C++, HTML, CSS, JavaScript, TypeScript, React, Node.js, Express.
- Design & Simulation: NI Multisim, Proteus, AutoCAD, Fusion 360, Eagle.
- Database: MySQL, PostgreSQL.
- Design & Multimedia: Adobe Photoshop, Adobe Lightroom, Canva, CapCut
- Microsoft Office: Word, Excel, PowerPoint.
- Languages: Indonesian (native), Malay, English (TOEIC Prediction 620/990 – working proficiency).